



SQL Functions Cheat Sheet



Essential Categories

We'll cover:

- Text & String Functions
- Date & Time Functions
- Aggregation & Math
- Window Functions
- Conditional Logic
- Data Cleanup & Conversion



STRING & TEXT FUNCTIONS

Function	Purpose	Example
UPPER() / LOWER()	Change text case	UPPER('hello') → 'HELLO'
TRIM()	Remove whitespace	TRIM(' data ') → 'data'
SUBSTRING()	Extract part of a string	SUBSTRING('ABCDEF', 1, 3) → 'ABC'
REPLACE()	Replace substrings	REPLACE('2023-01', '-', '/')
CONCAT()	Join strings	CONCAT(first_name, ' ', last_name)
LENGTH()	Count characters	LENGTH('banana') → 6



Ideal for cleaning names, building dynamic labels, or formatting fields for reports.



DATE & TIME FUNCTIONS

Function	Purpose	Example
CURRENT_DATE / NOW()	Get current date or timestamp	NOW() → 2025-07-28 19:54:00
EXTRACT()	Pull parts of a date (year, month, etc.)	EXTRACT(YEAR FROM date) → 2025
DATE_TRUNC()	Truncate date to unit (month, year)	DATE_TRUNC('month', date)
DATEDIFF()	Calculate difference between dates	DATEDIFF(day, start, end) → 30
DATE_ADD() / + INTERVAL	Add time to a date	date + INTERVAL '7 days'

🕒 Common in cohort analyses, time-based dashboards, and retention calculations.



AGGREGATION FUNCTIONS

Function	Purpose	Example
SUM()	Total numeric values	SUM(sales)
AVG()	Average	AVG(score)
COUNT()	Count rows or non-null values	COUNT(*), COUNT(user_id)
MIN() / MAX()	Lowest or highest value	MAX(order_total)
GROUP BY	Aggregate across categories	GROUP BY region

📊 Use these for metrics, KPIs, totals per dimension, or building summary tables.



WINDOW FUNCTIONS

Function	Purpose	Example
ROW_NUMBER()	Assign unique row IDs	ROW_NUMBER() OVER (PARTITION BY user ORDER BY date)
RANK() / DENSE_RANK()	Ranking with ties	RANK() OVER (ORDER BY profit DESC)
LEAD() / LAG()	Look ahead / behind row	LAG(sales, 1) OVER (ORDER BY month)
OVER()	Frame window for aggregate	SUM(sales) OVER (PARTITION BY region)

🧠 Perfect for running totals, period-over-period change, or comparative ranking.



CONDITIONAL LOGIC

Function	Purpose	Example
CASE WHEN	If-then logic	CASE WHEN sales > 100 THEN 'High' ELSE 'Low' END
IF()	Binary condition (MySQL)	IF(score > 80, 'Pass', 'Fail')
COALESCE()	Replace NULL with fallback	COALESCE(email, 'Not Provided')
ISNULL() / IFNULL()	Check for NULL	ISNULL(score)

⚙️ This is your toolbox for handling missing data, classification buckets, and customizing outputs.

🔧 DATA CLEANUP & TYPE CONVERSION

Function	Purpose	Example
CAST() / CONVERT()	Change data type	CAST(sales AS INT)
ISNUMERIC()	Check if value is numeric	ISNUMERIC(column)
NULLIF()	Return NULL if values match	NULLIF(price, 0)
DISTINCT	Remove duplicates	SELECT DISTINCT customer_id

💡 Use these during data ingestion or validation to prevent bad data downstream.

🧠 Analytical & Advanced Functions

Function	Purpose	Example
NTILE()	Divide rows into equal buckets	NTILE(4) OVER (ORDER BY sales)
PERCENT_RANK()	Relative rank as a percentage	PERCENT_RANK() OVER (ORDER BY score)
CUME_DIST()	Cumulative distribution	CUME_DIST() OVER (ORDER BY revenue)
FIRST_VALUE()	First value in a window	FIRST_VALUE(date) OVER (PARTITION BY user)
LAST_VALUE()	Last value in a window	LAST_VALUE(date) OVER (PARTITION BY user)

🔍 These are powerful for percentile analysis, customer segmentation, and performance benchmarking.

🧮 Mathematical & Statistical Functions

Function	Purpose	Example
ROUND()	Round numbers	ROUND(price, 2)
CEIL() / FLOOR()	Round up/down	CEIL(3.14) → 4
ABS()	Absolute value	ABS(-100) → 100
MOD()	Remainder	MOD(order_id, 2)
POWER()	Exponentiation	POWER(score, 2)

📊 Useful for financial modeling, scoring systems, and data normalization.

✿ Data Structuring & Transformation

Function	Purpose	Example
CASE	Conditional logic	CASE WHEN x > 10 THEN 'High'
CAST() / CONVERT()	Change data type	CAST(sales AS FLOAT)
UNION / UNION ALL	Combine result sets	SELECT A UNION SELECT B
INTERSECT / EXCEPT	Compare result sets	SELECT A INTERSECT SELECT B

💡 These help reshape data for reporting, dashboards, and ETL pipelines.

📅 Time Intelligence & Period Analysis

Function	Purpose	Example
TO_CHAR()	Format date/time	TO_CHAR(date, 'YYYY-MM-DD')
DATE_PART() / EXTRACT()	Pull date components	EXTRACT(MONTH FROM date)
DATEADD() / DATEDIFF()	Add or compare dates	DATEDIFF(day, start, end)
NOW() / CURRENT_TIMESTAMP	Current time	NOW()

📅 Crucial for time-series analysis, retention metrics, and forecasting.

🔍 Data Validation & Quality Checks

Function	Purpose	Example
ISNULL() / IFNULL()	Handle nulls	ISNULL(email, 'N/A')
NULLIF()	Avoid divide-by-zero	NULLIF(total, 0)
DISTINCT	Remove duplicates	SELECT DISTINCT user_id
CHECK()	Enforce constraints	CHECK(age > 0)

💡 These help ensure clean, reliable data for analysis and reporting.